Evolution of Quantum Computing: Theoretical and Innovation Management Implications for Emerging Quantum Industry

Mario Coccia, Saeed Roshani, and Melika Mosleh

***Abstract*—Quantum computing is a vital research field in science and technology. One of the fundamental questions hardly known is how quantum computing research is developing to support scientific advances and the evolution of path-breaking technologies for economic, industrial, and social change. This study confronts the question here by applying methods of computational scientometrics for publication analyses to explain the structure and evolution of quantum computing research and technologies over a 30-year period. Results reveal that the evolution of quantum computing from 1990 to 2020 has a considerable average increase of connectivity in the network (growth of degree centrality measure), a moderate increase of the average influence of nodes on the flow between nodes (little growth of betweenness centrality measure), and a little reduction of the easiest access of each node to all other nodes (closeness centrality measure). This evolutionary dynamics is due to the increase in size and complexity of the network in quantum computing research over time. This study also suggests that the network of quantum computing has a transition from hardware to software research that supports accelerated evolution of technological pathways in quantum image processing, quantum machine learning, and quantum sensors. Theoretical implications of this study show the morphological evolution of the network in quantum computing from a symmetric to an asymmetric shape driven by new inter-related research fields and emerging technological trajectories. Findings here suggest best practices of innovation management based on R&D investments in new technological directions of quantum computing having a high potential for growth and impact in science and markets.**

***Index Terms*—Innovation management, quantum algorithms, quantum computing (QC), quantum network, technological change, technological paradigm, technological trajectories.**

I. Introduction

QUANTUM computing (QC) research can improve information and communication technologies by allowing the transmission and manipulation of quantum bits (or qubits, which is the quantum mechanical analog of a classical bit) between remote locations [1]. QC is at the initial stage of technological evolution but has a high potential of generating innovations in quantum communication, quantum cryptography, quantum optics, etc. to support competitive advantage of firms and nations [2]–[6]. The evolution of QC needs R&D investments for creating a complete and functional quantum ecosystem—based on effective scientific and technological networks, reliable physical infrastructures, skilled human resources, etc.—that supports scientific advances and innovations in markets and society [7]–[11]. There is a vast literature in these research fields, however, the evolution of QC that develops path-breaking innovations is hardly known. This study confronts the problem here by developing a computational analysis based on publications over 1990–2020 period to explain the structure and evolution of QC research that support scientific and technological trajectories having implications for economic, industrial, corporate, and social change. This study can detect the main QC technologies, which may be major sources of competitive advantage for firms and nations to solve complex problems and satisfy new needs of people in society. In this context, the next section presents a theoretical framework to describe how maps of science, which we are going to create for QC, can explain the evolution of new technologies over the course of time.

II. Theoretical Framework

Scholars assert that the evolution of technologies is increasingly based on the interaction between technologies and scientific fields that generate co-evolutionary pathways of new technological trajectories [12]–[20]. In general, the evolution of technologies is driven by science that is often viewed as a self-organizing system with various scientific changes and interactions within and between social community of scholars [21]–[22]. The scientific development, underlying technological change, can be investigated with publications that are a main unit of analysis to show science maps of how scientific fields and technologies evolve over time [23]–[26]. In this context, Leydesdorff (2007) has developed a map of the whole set of journals, showing how centrality measures can clarify the environment of citations given by small sets of journals where citing is above a certain threshold [27]. Savov *et al.* [28] propose an approach of citation-based measures to identify breakthrough scientific

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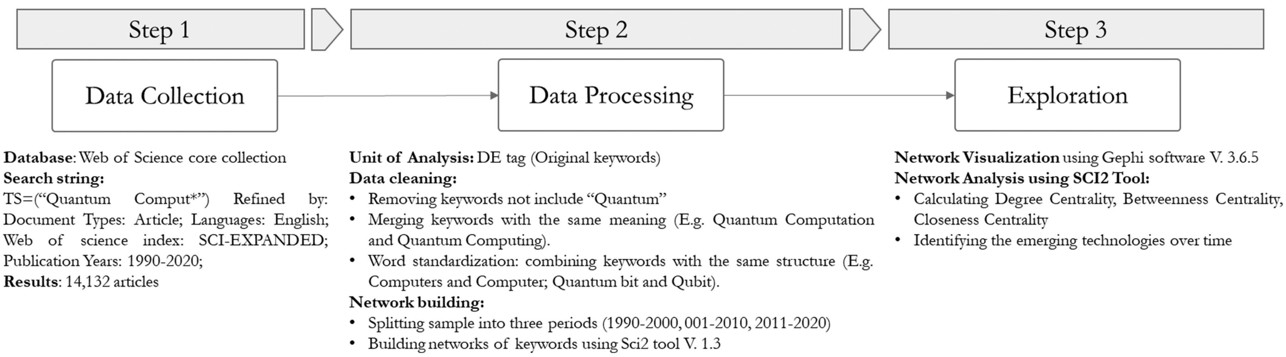

Fig. 1. Data processing procedure of this study.

papers driving science advances. Instead, Boyack *et al.* [29] study the accuracy of scientific maps using eight different intercitation and co-citation similarity metrics. Klavans and Boyack [30] identify a better measure of relatedness for mapping science because the measurement of the relatedness between bibliometric units (e.g., journals, words, etc.) provides critical aspects for structure and evolution of science. Relatedness measures have also a vital role in showing the relationship between data items in maps [30]. Small and Garfield [31] reveal that large and small areas of science are often arranged in center-periphery patterns. Small argues that the network of linkages from document to document and from discipline to discipline can show crossover fields and offer the possibility of exploring extended pathways of knowledge and new technological trajectories [32]. Boyack and colleagues also maintain that maps of science can identify major fields of research and emerging technologies, showing their size and interconnectedness [29]. The results of these studies can guide R&D investments and innovation strategy of firms and governments for supporting their competitive advantage in turbulent markets [33]–[37]. The evolution of emerging technologies, as anticipated, is also associated with underlying scientific development and change [33]. Cozzens and colleagues point out that emerging technologies present new opportunities for national competitive advantage [38]–[39]. Manifold techniques have been developed in scientometrics to detect and analyze new domains in science and technology [24]–[43]. These methods are based on large datasets and computational approaches based on complex indicators for detecting new technological trajectories using data from publications and/or patents [44]–[46]. In particular, quantitative approaches based on bibliometric data of journals, compared to patents, can capture information of innovations early in their technological development ([26], [38], [39]). In this context, Deshmukh and Mulay [47] show that the maximum number of publications in the field of QC is in physics, astronomy, and computer science. However, studies of scientometrics concerning evolutionary networks of QC associated with new technologies are scarce in the literature. This study endeavors to create and analyze the mapping of QC over a 30-year time frame to detect different stages of the structure of network and the evolution of QC with main technological trajectories. The next section presents the methodology to develop the purpose of this study.

III. Material and Method

Fig. 1 shows the methodological approach of the present study based on: (1) data collection, (2) data processing, and (3) exploration of networks in QC.

A. Data Collection and Sample

We used the Web of Science-WOS core collection database for extracting the articles related to QC [48]. To collect the most relevant articles in this field of research, we searched "Quantum Comput*" through the topics of articles. Afterward, the results were limited to document type = ("Articles"), language = ("English"), Web of Science index = ("Science Citation Index Expanded") to gather helpful data. Data are over the 1990–2020 period. The sample contains 14,132 articles split into three timespans, given by 1990–2000, 2001–2010, and 2011–2020.

B. Data Processing and Analysis

First, we used original keywords of articles (DEs) tag as the basis for detection of technologies related to QC. Among Web of Science's special tags, "DE" is one that is related to the author's keywords and can be used to build a keyword co-occurrence network. Web of Science provides another source of keywords that are related to the keyword plus (IDs) that Web of Science database assigns to the documents [48]. Hence, in this article, the units of analysis are the original keywords (DEs) of articles provided by authors to find the QC research fields and technologies and to create the interconnection network among nodes. Second, we have done the cleaning of data for constructing the networks. We kept only the key phrases combined with the word "quantum." We also merged all keywords with the same meaning written in different structures, including abbreviation, plural, and singular forms, and we put the gerund and noun forms together and considered them as one single node. Additionally, we eliminated the isolated nodes, which have no connections with other nodes in the graph.

C. Creation and Exploration of Networks

Finally, we generated co-occurrences networks by SCI2 Tool V.1.3 software [49] and imported them as a GraphMl format file into Gephi software version 3.6.5 [50] to visualize and analyze

TABLE I
COMPARISON OF NETWORK MEASURES AND INDICATORS OVER THREE PERIODS

	1990-2000	**2001-2010**	*Growth from (1990-2000) To (2001-2010)*	**2011-2020**	*Growth from (2001-2010) To (2011-2020)*
Total Number of Articles	668	5,224	6.82	9,189	0.76
Number of Nodes	87	477	4.48	1,054	1.21
Number of links or edges	140	793	4.66	2,244	1.83
Graph Density	0.037	0.007	-0.81	0.004	–0.43
Avg. Path Length	2.332	2.748	0.18	2.859	0.04

Note: Nodes represent vertices in a graph; links (or edges) are connections between nodes (or vertices) of the network. Graph density is the maximum number of existed edges divided by the number of edges; Avg. path length is the number of steps in average, through the shortest routes between all joints of network nodes.

the networks for each period under study (from 1990 to 2000, 2001 to 2010, and 2011 to 2020). The structure of networks is based on nodes and edges. In the keywords co-occurrences network, each *node* represents subtopics related to QC research and technologies. An *edge* represents a linkage between two keywords when they appear in the same document. The *thickness* of each edge indicates the frequency of appearance of a specific node with other connected nodes in documents, i.e., when the edge of nodes AB is thicker than the edge of nodes AC, the co-occurrences frequency between node A and node B is higher than the frequency of the link between node A and node C.

Moreover, SCI2 Tool is applied to calculate metrics of these networks, such as degree centrality (DC), betweenness centrality (BC), and closeness centrality (CC) measures to explore the evolutionary behavior of different nodes within the network of QC over time. In particular,

a) DC is the number of connected edges to a node [51]. DC indicates the number of connections (connectivity) in the network of keywords.
b) BC indicates the amount of influence or control a node has over the flow between nodes and networks (similar to a bridge). This index represents the importance of a node in making the connection between other network nodes and is responsible for sustaining the network integration [52]. We implemented BC measures to show the bridge nodes that facilitate the linkage of entities.
c) A node's CC is an indicator of a network centrality, defined as the number of links needed to connect each node in the network with all the other nodes in the network or the average number of links required to reach all other nodes in the network from a node in the network [53].

These measures provide comprehensive information for a comparative analysis of the evolutionary pathways of networks in QC over time [54]–[55].

IV. RESULTS

Table I shows that the interconnection of QC-related words is structured in the 1990–2000 period through 87 nodes and 140 edges based on 668 articles (cf., Fig. 2). The graph density of this network is 0.037, which respectively shows a considerable integration belongs to the network. Between 2001 and 2010, based on 5,224 publications, the number of nodes increased by 4.48 times (it is 477), and the number of edges increased by 4.66 times (the number is 793) with also a reduction in graph density, which is 0.007 in the second period under study (see Fig. 3). In the last period, 2011–2020, the number of nodes increased by 1.21 times compared to the previous period. This graph has a density of 0.004, which is less than the graph density of 2001–2010 network (see Fig. 4). The average path length of the last period graph is 2.859, a higher value than the two other periods. Interestingly, the interconnection network contains 2,244 edges between 2011 and 2020, with an increase of 1.83 times over the preceding decade.

A. *Evolution of Network in QC Research from 1990 to 2000*

In the first period, Table II and Fig. 2 show that quantum computer and quantum Turing machines have the highest DC. These findings show that these nodes have a main role in structuring the graph and establishing new linkages with different nodes in that period, which caused the highest level of connection score. Meanwhile, QC (item) with the BC of 0.486 and quantum computer with 0.118 established essential bridges among other network nodes to facilitate the interaction and coevolution between different topics [16]. Furthermore, Fig. 2 shows that QC node has the intertwined connection with the quantum algorithm, quantum information, and quantum dot. Instead, quantum turing machine with a high centrality degree has a strong connection with quantum polynomial time. In this network, quantum optics, quantum searching, quantum lattice gas, quantum quantitates,

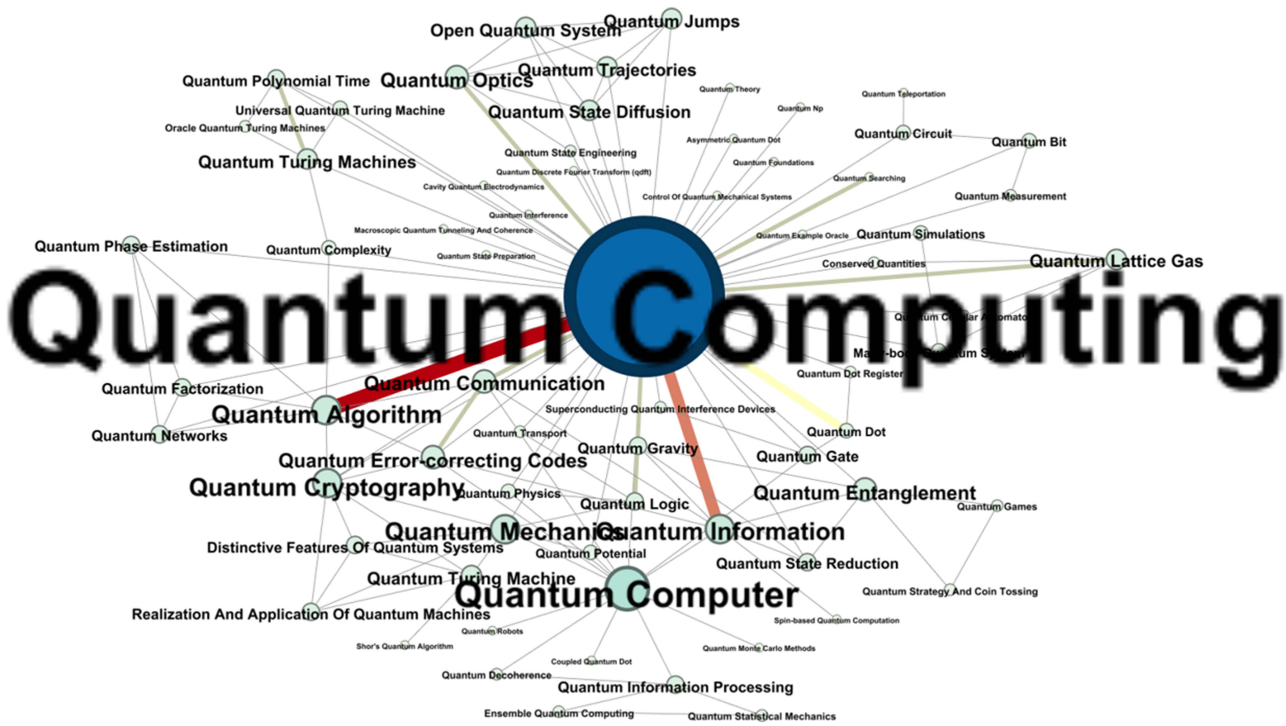

Fig. 2. Evolution of interconnection in the QC-related words, 1990–2000 period.

quantum logic, and quantum communication also illustrate a significant connection with QC node.

B. *Evolution of Network in QC Research from 2001 to 2010*

In the 2001–2010 period, the quantum network has a great expansion. In this regard, Table II shows that quantum information and quantum algorithms grasped the highest level of DC after QC and quantum computer. Moreover, the node of the quantum turing machine is eliminated from the top nodes. As far as BC is concerned, quantum dot with a value of 0.162 had a significant role in bridging the network. This node has BC value that is more than twice of the quantum computer, while its DC value is less than a half of the quantum computer. According to Table II (2001–2010 period) and Fig. 3, there is a strong and established linkage between nodes of quantum information and QC. Quantum information in the first period utilized its initial connections with QC to grow faster; after ten years, it is one of the central nodes with the highest value of DC. Moreover, Fig. 3 shows that quantum gate and quantum dot have created new strong linkages with the node of quantum computer, creating potential characteristics of coevolution over the course of time. In addition, Fig. 3 shows an interesting triangle connection pattern between quantum optics, quantum entanglement, and quantum communications, which is a starting point of intensive interaction and rapid coevolution pathways between these nodes in the evolution of QC network.

C. *Evolution of Network in QC Research from 2011 to 2020*

In the period from 2011 to 2020, Table II and Fig. 4 show that quantum algorithm has been experiencing a considerable growth in the DC measure. In this period, it has expanded even more than quantum computer and can be considered the second highest DC node in the interconnection network of QC. Moreover, during this period, quantum information and quantum correcting-error code have a higher DC value than quantum computer: this result shows the growing importance of software development in QC. In fact, these results also illustrate a transition of QC research from hardware to software aspects in evolutionary development. Although quantum algorithm and quantum information have the highest DC and expand their connections through the evolution of network, they still have a weaker bridging role compared to quantum computer, such that these nodes cannot establish their essential role to facilitate the connections among other nodes. This finding suggests that they are in the initial phase of technological growth, but their BC has potential factors for growth in a not-too-distant future; especially, quantum algorithm and quantum information will play an essential role in establishing an integrated evolution within network of QC research. Moreover, in this period, quantum processing and quantum circuit are new nodes that started their connection with QC. This scientific change in linkages and nodes shows a continuous transition and evolution of this scientific domain.

D. *General Evolution of the Network in QC Research and Emerging Technologies for Innovative Applications in Markets*

An in-depth technology analysis within the network of QC shows that in the period 1990—2000, the highest connectivity is due to nodes of quantum turing machine (DC $=$ 9) and quantum information (DC $=$ 8), in addition, of course to main

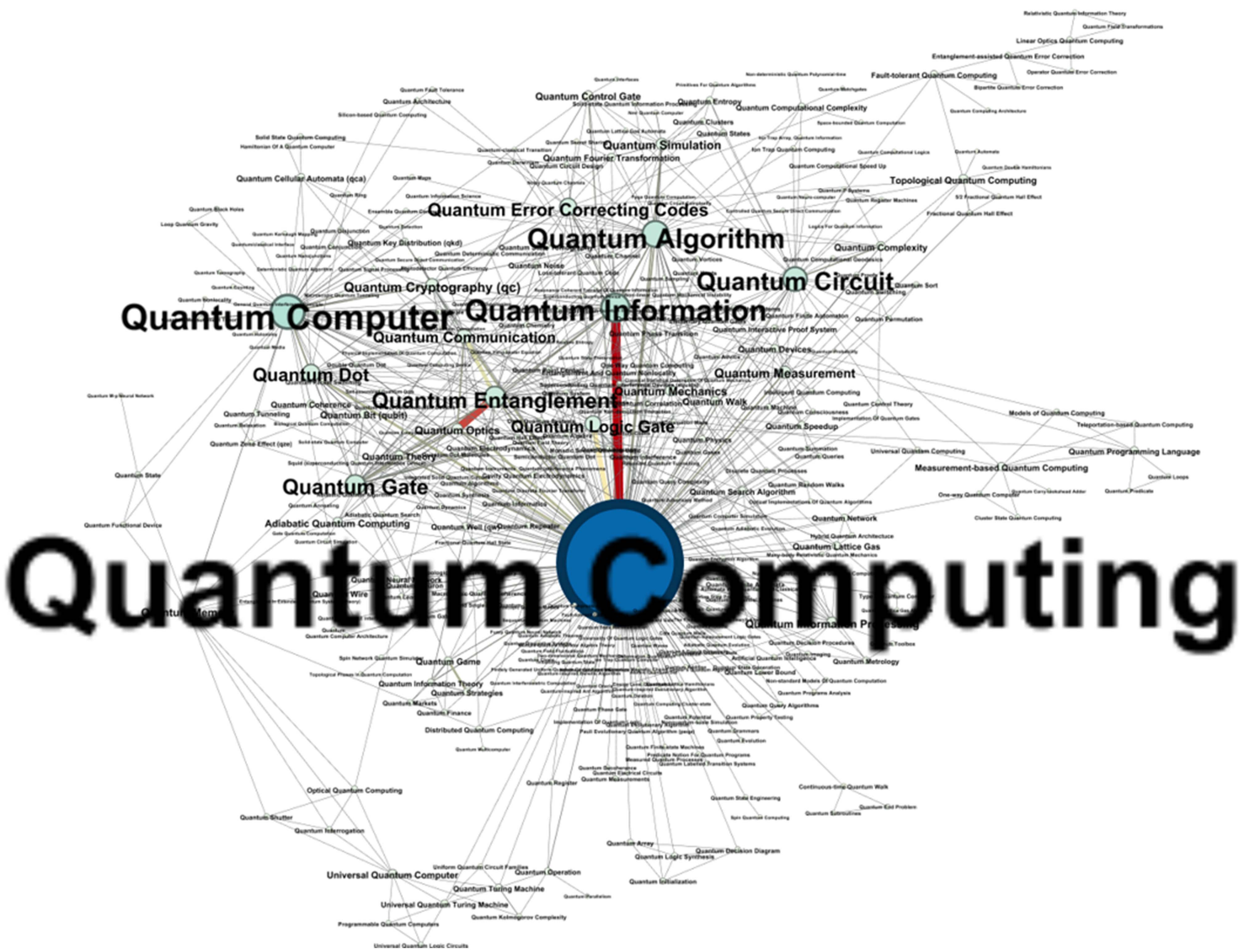

Fig. 3. Evolution of interconnection in the QC-related words, 2001–2010 period.

nodes of QC and quantum computer. In 2001–2010 period, the highest connectivity is driven by quantum information with a considerably increase of DC = 45 and quantum algorithm having DC = 41. In the last period (2011–2020), the acceleration of connectivity is due to quantum algorithm (DC = 102), quantum information with DC = 90, and the new entry of quantum error-correcting code (DC = 65). These evolutionary aspects of QC research suggest, as said, a transition from hardware to software characteristics of the science dynamics. In this scientific and technological evolution, the nodes having a high influence or control over the flow between nodes in the network are, in 1990–2000 period, the QC, quantum computer, quantum entanglement, and quantum information processing, whereas in the 2011–2020 period, they are QC, quantum computer, and new entry of quantum algorithm and quantum information. Finally, CC suggests that nodes having the easiest access to all other nodes in the network, over 1990–2000 period, are quantum lattice gas, quantum jumps, quantum state diffusion, and quantum trajectories; over 2001–2010 period, main nodes are QC, quantum computer, quantum information, and quantum algorithm; finally, in the last period of 2011–2020, we have quantum mechanics, quantum annealing, quantum image processing, and quantum dot with a high CC measure. A general technology analysis of the evolution of the top 20 fields and technologies in QC from 1990 to 2020 shows a considerable average increase of connectivity with a growth of 7.43 (DC), a moderate increase of 0.12 (BC) about the average influence or control of nodes on the flow between nodes within network, whereas CC measure has a very low reduction—0.06 (the average easiest access of nodes to all other nodes in technological network for top 20 QC research and technologies under study). This evolutionary dynamics is due to the increase in size and complexity of the technological and scientific network in QC (see Table III).

The emerging technologies in QC from 2001 to 2010 are (see Table IV): quantum memory, adiabatic QC, topological QC, quantum walk, and quantum wire. Furthermore, over 2011-2020, the emerging technologies are quantum image processing, quantum machine learning, and blind QC. This result suggests the potential transfer of QC research in new technologies for innovative applications, such as in quantum image processing for creating quantum images because this new technology has high technical performance in terms of computing speed, low storage requirements, and high security [56]. Other emerging technologies in QC research are quantum machine learning based on the interaction of machine learning programs and quantum algorithms [57] and blind QC, such that quantum

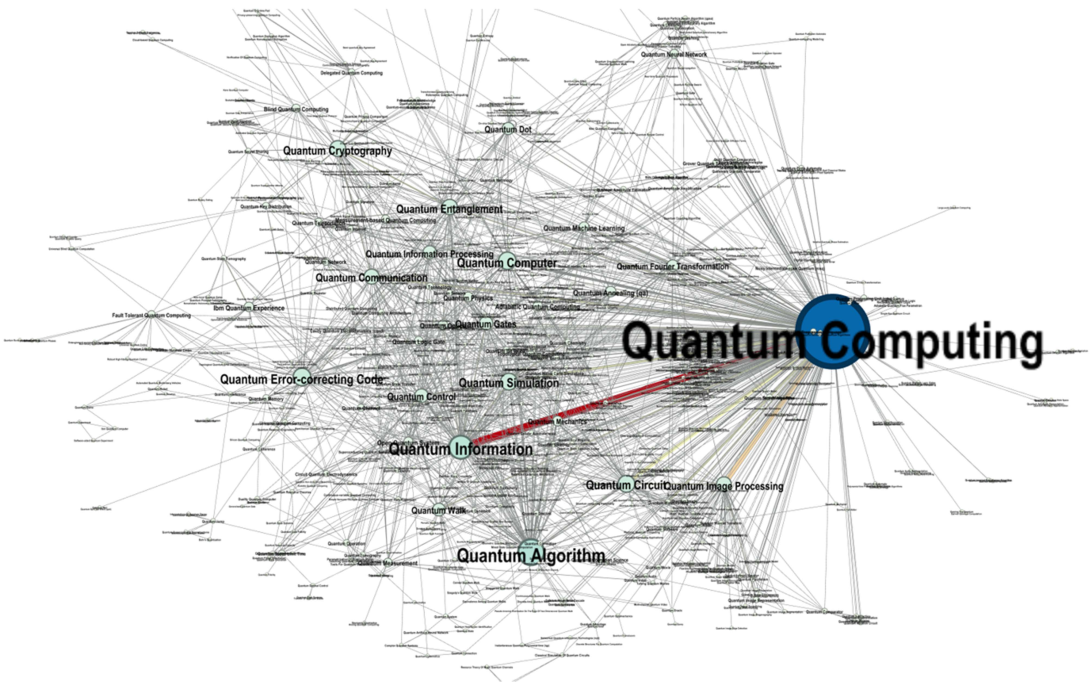

Fig. 4. Evolution of interconnection in the QC-related words, 2011–2020 period.

servers do not have full information about activities that users are computing for high security of tasks. In this innovative activity, a new direction is due to delegated QC in which users operate on server's resources for assessing a complex circuit without leaking any information about the input to the server. Another emerging technology is circuit quantum electrodynamics to analyze fundamental interaction between light and matter (quantum optics), quantum internet for innovative communications by teleportation, and quantum sensors that detect variations in microgravity for altering elements of the nature at the submolecular level with main applications in microscopy, positioning systems, mineral exploration, seismology, optical quantum sensors for quantum lithography, etc. [58]–[59].

V. Discussion With Theoretical and Innovation Management Implications

The evolution of QC network over the last three decades is unparalleled [60]. Scholars have analyzed quantum research and suggested main technological pathways that, of course, are not exhaustive: (i) quantum information; (ii) quantum sensing and imaging; (iii) quantum communication and cryptography; and (iv) quantum computation [61]–[63]. In this research stream, the study here shows the scientific and technological evolution of network in QC over a 30-year period (from 1990 to 2020). Results show that the network of QC is rapidly growing from 2001 to 2020. In particular, this study suggests that the research fields of QC are in continuous evolution because of recent advances in physics, optics, and computer science. In fact, in the period 2001–2010, a lot of new technologies emerged, developed, and connected to other ones, such as quantum memory, adiabatic QC, topological QC, quantum walk, quantum wire, quantum programming language, and quantum cellular automata. In addition, emerging technologies over 2011–2020 period are focused on quantum image processing, quantum machine learning, blind QC, delegated QC, circuit quantum electrodynamics, and quantum image representation [64]. These findings here reveal that the dynamics of QC research evolves with endogenous processes of high connectivity between research fields and technologies that increase the size and complexity of the quantum computing network supporting research fields, scientific and technological trajectories [22].

A. *Principal Theoretical Implications from the Evolution of QC network*

These results suggest some properties of the scientific and technological change of the network of QC research that can support general principles for the evolution of science and technology.

First, QC research co-evolves with complex interactions driven by three evolutionary characteristics: a) a high *connectivity* between nodes of research fields and technologies; b) *moderate growth of the average influence* of nodes on the flow between nodes within network; c) *a reduction about the average easiest access of nodes* to all other nodes because of a larger size and complexity of QC network over the course of time.

Second, some research fields and technologies in QC network have a critical position, playing a role of *master entity* for connections of manifold research fields and technologies, such as quantum algorithm and information.

Third, QC network is generating, during the co-evolution of research fields, new technological trajectories from a process

TABLE II
NETWORK MEASURES IN QUANTUM COMPUTING RESEARCH AND TECHNOLOGIES FROM 1990 TO 2020

1990-2000				2001-2010				2011-2020			
Label	DC	BC	CC	Label	DC	BC	CC	Label	DC	BC	CC
Quantum Computing	51	0.486	0.802	Quantum Computing	211	0.380	0.681	Quantum Computing	412	0.369	0.661
Quantum Computer	13	0.118	0.532	Quantum Computer	55	0.070	0.505	Quantum Algorithm	102	0.039	0.486
Quantum Turing Machine	9	0.032	0.485	Quantum Information	45	0.040	0.495	Quantum Information	90	0.030	0.488
Quantum Information	8	0.024	0.507	Quantum Algorithm	41	0.033	0.491	Quantum Error-correcting Code	65	0.020	0.463
Quantum Algorithm	8	0.002	0.474	Quantum Circuit	37	0.036	0.484	Quantum Computer	64	0.193	0.458
Quantum Cryptography	7	0.006	0.477	Quantum Entanglement	31	0.011	0.469	Quantum Circuit	61	0.015	0.462
Quantum Mechanics	7	0.012	0.503	Quantum Gate	30	0.014	0.474	Quantum Simulation	61	0.022	0.456
Quantum Communication	6	0.001	0.496	Quantum Dot	25	0.162	0.461	Quantum Cryptography	57	0.017	0.453
Quantum Error-correcting Codes	6	0.001	0.496	Quantum Error Correcting Codes	24	0.024	0.474	Quantum Entanglement	55	0.016	0.462
Quantum Entanglement	6	0.034	0.474	Quantum Logic Gate	23	0.020	0.457	Quantum Communication	53	0.009	0.466
Quantum Optics	6	0.0005	0.464	Quantum Communication	19	0.009	0.454	Quantum Image Processing	52	0.014	0.439
Open Quantum System	5	0.0	0.460	Quantum Cryptography	16	0.006	0.448	Quantum Control	47	0.012	0.455
Quantum Lattice Gas	5	0.0006	0.460	Quantum Measurement	15	0.004	0.439	Quantum Gates	46	0.011	0.459
Quantum Jumps	5	0.0	0.460	Quantum Simulation	14	0.007	0.440	Quantum Information Processing	45	0.015	0.449
Quantum State Diffusion	5	0.0	0.460	Quantum Physics	14	0.003	0.433	Quantum Walk	43	0.014	0.447
Quantum Trajectories	5	0.0	0.460	Quantum Control Gate	13	0.017	0.459	Quantum Dot	40	0.019	0.440
Quantum Factorization	5	0.0	0.457	Quantum Mechanics	13	0.003	0.446	Quantum Fourier Transformation	38	0.011	0.443
Quantum Gate	4	0.002	0.481	Quantum Memory	12	0.012	0.435	Quantum Annealing	33	0.007	0.437
Quantum Information Processing	4	0.034	0.359	Quantum Information Processing	12	0.004	0.435	Quantum Machine Learning	31	0.006	0.444
Quantum Gravity	4	0.0	0.467	Quantum Optics	11	0.0004	0.427	Quantum Mechanics	30	0.003	0.436

Note: Degree Centrality (DC) indicates number of connections (connectivity); Betweenness Centrality (BC) indicates the amount of influence or control a node has over the flow between nodes and networks (similar to a bridge); Closeness Centrality (CC) indicates the easiest access to all other nodes in a network or sub-network (shortest distance from nodes).

of specialization in science, such as quantum machine learning, blind QC, quantum sensor, quantum image processing, etc.

Fourth, the morphological evolution of network in QC research is changing from spheroid shape in the initial stage of evolution to an irregular shape in the stage of growth over 2011–2020 period (see Fig. 5).

- *Spheroid type* (1990–2000 period) is a scientific and technological network in the initial phase of evolution having a symmetric shape with nodes and mutual interconnexions sparse.
- *Irregular type* (2011–2020 period) is a scientific and technological network in the growing phase of evolution with an asymmetric shape having dense nodes, high mutual interconnexions, and high connectivity. Inside the scientific and technological network, the evolution is based on endogenous processes given by: *merger* when two or

TABLE III
PATTERN OF THE EVOLUTION OF QUANTUM COMPUTING RESEARCH FROM 1990 TO 2020, BASED ON TOP 20 FIELDS AND TECHNOLOGIES OF TABLE II

Period	Degree Centrality (DC)	Betweenness Centrality (BC)	Closeness Centrality (CC)
1990-2000	8.45	0.038	0.496
2001-2010	33.05	0.043	0.470
2011-2020	71.25	0.042	0.465
% growth of 2001-2010 on 1990-2000 period	2.91	0.14	–0.05
% growth of 2011-2020 on 1990-2000 period	**7.43**	**0.12**	**–0.06**
% growth of 2011-2020 on 2001-2010 period	1.16	-0.016	–0.011

Note: Degree Centrality (DC) indicates number of connections (connectivity); Betweenness Centrality (BC) indicates the amount of influence or control a node has over the flow between nodes and networks (similar to a bridge); Closeness Centrality (CC) indicates the easiest access to all other nodes in a network or subnetwork (shortest distance from nodes).

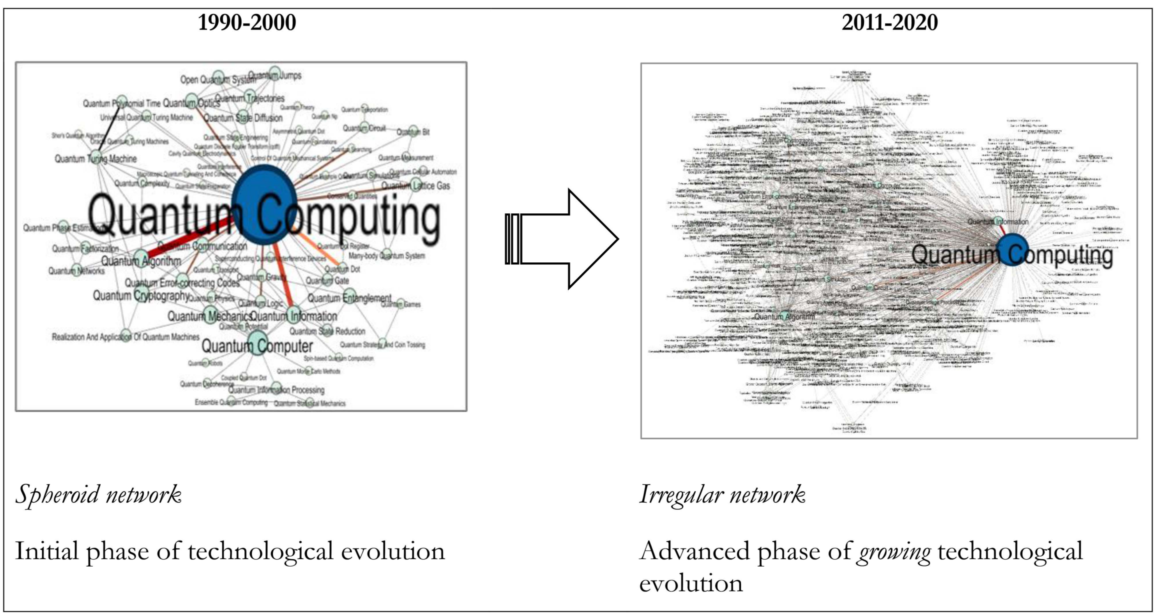

Fig. 5. Morphological evolution of network in QC research.

more nodes (technologies or scientific fields) combine their elements or subsystems to generate a new large system; *interacting pair* when two nodes interact with a beneficial coevolution having aspects of mutualism and symbiosis over time; finally, *splitting* when a node grows for the accumulation of scientific research and causes a split off of some subsystems, generating two or more new nodes (independent systems of new research fields and technologies).

B. Principal Implications of Innovation Policy and Innovation Management for a Quantum Industry

The evolutionary pathways in QC have main implications for the management of technologies and innovations to support decisions of R&D investments of firms and governments for competitive advantage in markets. In fact, policymakers and R&D managers know that financial resources can be an accelerator factor for progress and diffusion of science and technology to support the scientific and technological development in society [65]. This study shows critical research fields and technologies in QC that are growing with a higher DC measure; R&D management can allocate economic resources towards these research fields and technologies (e.g., quantum memory, quantum image processing, quantum machine learning, etc.) to support scientific and technological development that create a background for competitive advantage in markets. In addition, policymakers and funding agencies can use these findings here for making efficient decisions of sponsoring specific research fields and technologies in QC to foster technology transfer having fruitful effects for economic growth and social change in nations.

TABLE IV
TOP 20 EMERGING QUANTUM COMPUTING TECHNOLOGIES FROM 2001 TO 2020

	2001-2010		**2011-2020**	
Rank	Label	Degree Centrality	Label	Degree Centrality
1	Quantum Memory	12	Quantum Image Processing	52
2	Adiabatic Quantum Computing	10	Quantum Machine Learning	31
3	Topological Quantum Computing	9	Blind Quantum Computing	23
4	Quantum Walk	9	Delegated Quantum Computing	16
5	Quantum Wire	7	Circuit Quantum Electrodynamics	14
6	Quantum Programming Language	7	Quantum Image Representation	13
7	Quantum Cellular Automata	7	Quantum Internet	13
8	Quantum Entropy	6	Quantum Processing Unit	13
9	Quantum computing Complexity	6	Quantum Sensor	13
10	Quantum Well	6	Noisy Intermediate-scale Quantum	12
11	Quantum Interactive Proof System	6	Quantum Adder	12
12	Quantum Neural Network	6	Quantum Amplitude Estimation	12
13	Fault-tolerant Quantum Computing	6	Quantum Comparator	12
14	Quantum Noise	6	Quantum Filtering	12
15	Quantum Key Distribution	6	Quantum Monte Carlo Simulations	12
16	Quantum Phase Transition	5	Quantum Software	12
17	Optical Quantum Computing	5	Quantum Watermarking	12
18	Distributed Quantum Computing	5	Quantum Discord	11
19	Quantum Metrology	5	Variational Quantum Algorithm	11
20	Quantum Neuron	5	Quantum Amplitude Amplification	10

VI. CONCLUSION

The evolution of QC research shows that from 2001 to 2020 the number of nodes has a continuous growth, increasing the connectivity between nodes and increasing nodes that control the flow between nodes. The evolution of network in QC shows that new research fields and technologies emerge, such as quantum image processing, quantum machine learning, etc. Moreover, the morphological evolution of network in QC has changing shape from a spheroid typology in the initial phase of evolution (1990–2000) with sparse nodes and lower interconnections to an irregular type in the phase of growth with a high density of nodes and intensive interactions between nodes that generate an asymmetric shape with large dominant nodes, such as quantum information, quantum algorithm, etc., generating the continuous expansion of the domain of this research field in the universe of science (see Fig. 5). Results also reveal that network of QC research is evolving with a transition from hardware to software aspects.

These conclusions are, of course, tentative. Although this study has provided some interesting, albeit preliminary results, as many other bibliometric studies, it has several limitations. First, the precision of the search queries is affected by ambivalent meanings in QC, such as information, computing, computer, etc. Second limitation of this study is that the sources under study may only capture certain aspects of the ongoing evolutionary dynamics of QC research. Third, there are multiple confounding factors that could have an important role in the dynamics of QC research to be further investigated, such as multiple discoveries [66], R&D investments, role of scientific institutions, collaboration intensity, intellectual property rights, etc. [67]–[73]. Fourth, the network of QC research changes its borders during the evolution of science, and as a consequence, the morphology of network during the evolution, generating an irregular and complex shape, such that the identification of stable technological trajectories and scientific fields over the course of time is a difficult exercise.

To conclude, future research should consider new data when available and apply new approaches to reinforce proposed results here. The future development of this study is also directed to design indices of technometrics based on measures of betweenness, closeness, and DC of networks to assess technological and scientific change, to predict the emergence of new technological trajectories, as well as to support further implications for the management of technology and R&D management. Despite these limitations, the results presented here clearly illustrate the main evolutionary paths of QC research that are increasingly based on growing connectivity between technologies and research fields, but a further detailed examination is needed for explaining the driving factors of the technological evolution and supporting an appropriate strategic management of innovation that supports a new quantum industry for the competitive advantage of firms and nations in turbulent markets.

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